

lava 7, 24, 25, 30, 31
 leaves 62–63
 Lemaître, Georges 11
 lemon sharks 88–89
 leopard seals 109
 levees 37
 life
 conditions for 15
 on Earth 42–45, 136–137
 life cycles 6–7
 lightning 35
 lilies 59
 lions 121
 litters 88, 126
 live young
 dolphins 116, 117
 mammals 44
 reptiles 105, 106, 107
 sea horses 93
 sharks 88–89
 lizards 106–107
 locusts 121
 lunar seas 16

M

magma 24, 30, 31
 mammals 7, 44, 45,
 116–119, 122–131,
 132, 135
 mammary glands 118
 mangroves 86, 88–89
 Mars 14
 marshes 37, 86
 marsupials 118–119
 mass extinction 97
 mating
 amphibians 94
 birds 108, 111, 112,
 114–115
 dinosaurs 99
 earthworms 75
 fish 88, 92
 insects 83, 84
 mammals 116, 118, 122,
 124–125, 127, 129, 130
 octopuses 70–71
 pterosaurs 100
 reptiles 102–103, 105,
 106
 spiders 76, 85

mayflies 83
 meanders 36
 meerkats 120, 126
 melting 39, 41, 136
 Mercury 14, 15
 Mesozoic era 96
 metamorphic rock 24, 25
 metamorphosis 80, 81, 82,
 83, 95
 meteorites 18
 mice, house 60
 microbes 42
 Mid-Atlantic Ridge 23
 middle age 135
 migrations
 humans 133
 lemon sharks 88
 Monarch butterflies 81
 Serengeti 122, 123
 sockeye salmon 91
 swallows 112, 113
 milk 44, 116, 118, 123,
 124, 125
 Milky Way 11
 mimic octopuses 71
 mimosa 65
 minerals 25, 26, 27
 mirror orchids 59
 Mississippi River delta 37
 moles 126
 molting 82, 85, 109
 monarch butterflies 80–81
 moon 16–17, 29
 moons 11, 14
 Morganucodon 44
 mosquitoes 83
 mosses 7, 43, 50
 moths 81, 129
 mountains 6, 22, 28–29
 mudskippers 89
 mushrooms 52–53
 musk oxen 117
 mycelium 53

N

naked mole rats 126–127
 Neptune 14
 nests
 birds 110, 112, 114, 131
 dinosaurs 99

fish 90
 insects 78–79, 127
 mammals 126, 130, 131
 pterosaurs 100
 reptiles 102, 103, 104,
 105
 neutrons 10
 Nile crocodiles 105
 nipples 117, 118, 119
 nucleus 46, 47
 nurseries 88, 116, 117
 nursery-web spiders 76
 nutrients 6, 53, 60, 64, 66,
 84, 85, 120
 nuts 62
 nuzzling 122
 nymphs 82, 83, 85

O

oak trees 62–63
 oases 100
 oceans 22, 23
 carbon cycle 40
 fish 88–93
 life on Earth 42, 43, 44
 life in water 867
 marine mountains 29
 rock cycle 25
 seed dispersal 56
 turtles 102–103
 water cycle 32–33
 octopuses 70–71
 old age 135
 olive ridley turtles 102
 oothecas 85
 opossums 119
 orangutans 130–131, 132
 orbicular batfish 89
 orbits 14–15, 17, 19
 orchids 58–59
 Oryctodromeus 98–99
 oxbow lakes 37
 oxpeckers 123

P

Pacific Ocean 91
 paleontologists 26, 27
 palm oil 57
 Pangaea 22, 23

paper wasps 79
 parasites 66, 123
 parenting
 amphibians 95
 birds 109, 110, 111, 112,
 124
 dinosaurs 98
 fish 92–93
 humans 134–135
 male 92, 93, 94,
 108–109, 110
 mammals 109, 111,
 116–119, 123, 124,
 126, 129, 130–131
 pterosaurs 100, 101
 penguins 38, 39, 108–109
 peregrine falcons 115
 pesticides 137
 pheromones 84
 photosynthesis 40, 62
 pigeons 124
 pillow lavas 30
 pitcher plants 65
 planetary nebulae 12, 13
 planets 6, 11, 14–15, 29
 plankton 71, 90, 91
 plants 6, 7, 48–67
 carbon cycle 40
 food chains 120, 121
 human impact 136, 137
 life on Earth 43, 44
 water cycle 32
 plastic waste 136
 Plateosaurus 96
 platypuses 45
 pods, dolphins 116, 117
 polar animals 87
 polar bears 124–125, 136
 pollination 6, 7, 44, 50, 51,
 54, 56, 58–59, 64
 pollution 73, 136
 polyps 73, 86
 Pompeii 31
 potter wasps 79
 pouches
 frigate birds 114
 kangaroos 118, 119
 penguins 108, 109
 sea horses 92, 93
 prairie voles 111
 praying mantises 84–85